

On Waring’s Problem and the Dickson–Pillai Condition: Padé Approximations, Diophantine Obstructions, and Computational Frontiers

Emil Kerimov*

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Abstract

In 1936, L. E. Dickson and S. S. Pillai established that the exact value of $g(k)$ in Waring’s problem is given by $g(k) = 2^k + \lfloor (3/2)^k \rfloor - 2$, provided that the inequality $r_k + q_k \leq 2^k$ (where $3^k = q_k 2^k + r_k$) holds for all natural numbers $k \geq 1$. While Kurt Mahler (1957) proved via p -adic Diophantine approximation that this condition can fail for at most finitely many k , Mahler’s theorem is ineffective. Effective methods (Baker, Beukers 1981, Dubickas 2006) established $\|(3/2)^k\| > 2^{-ck}$ with $c \approx 0.999999$, failing to reach the requisite barrier $c \leq \log_2(4/3) \approx 0.415037$.

In this paper, we explore the theoretical landscape of higher-dimensional Hermite–Padé approximants around smooth rational bases such as $1 - 1/81 = 80/81 = 5 \cdot (2/3)^4$. We analyze the mechanism by which simultaneous linear forms distribute asymptotic logarithmic height across independent relations ($c_{\text{eff}} \approx 0.3999$). We then identify and formalize the three critical mathematical obstructions that prevent a naive extension to an unconditional proof: (1) prime-factor leakage outside the base set $S = \{2, 3, 5\}$ (e.g., $r_8 = 161 = 7 \times 23$), which invalidates elementary S -unit product formula closures; (2) the one-dimensional trajectory constraint of exponential orbits $(3/2)^k$ under multi-dimensional subspace transference; and (3) non-asymptotic height growth dominating small- k bounds. Finally, we present high-throughput parallel verification results extending the zero-exception frontier.

1 Introduction and Historical Background

In 1770, Edward Waring posed his celebrated problem: does there exist for every natural number $k \geq 1$ a positive integer $g(k)$ such that every positive integer is the sum of at most $g(k)$ positive k -th powers? Following David Hilbert’s 1909 existence proof, Leonard Eugene Dickson [1] and Subbayya Sivasankaranarayana Pillai [2] proved independently in 1936 that:

$$g(k) = 2^k + \lfloor (3/2)^k \rfloor - 2, \tag{1.1}$$

subject to the condition that the remainder $r_k = 3^k \bmod 2^k$ and quotient $q_k = \lfloor (3/2)^k \rfloor$ satisfy:

$$r_k + q_k \leq 2^k \quad \text{for all } k \geq 1. \tag{1.2}$$

*Independent Mathematics Researcher. Correspondence: emil.kerimov93@gmail.com

Dividing by 2^k , condition (1.2) can be expressed in terms of the distance from $(3/2)^k$ to the nearest integer:

$$1 - \{(3/2)^k\} \geq (3/4)^k \cdot \left(\frac{1 - (2/3)^k}{1 - 2^{-k}} \right) \approx (3/4)^k. \quad (1.3)$$

1.1 Theoretical Landmarks and the Exponent Gap

1. **Mahler's Ineffective Finiteness (1957):** Using Ridout's p -adic extension of Roth's theorem, Kurt Mahler [3] proved that (1.3) has at most finitely many integer exceptions. However, Roth's method is inherently ineffective, providing no computable threshold K_0 .
2. **Beukers' Effective Bound (1981):** Frits Beukers [4] used Padé approximations to the algebraic function $(1 - 1/9)^{1/2} = \frac{2\sqrt{2}}{3}$ to establish the effective lower bound:

$$\|(3/2)^k\| > 2^{-ck} \quad \text{for all } k \geq K_0, \quad (1.4)$$

with $c = 0.999999$. This was refined by Dubickas [6], but remains bounded away from the target barrier:

$$c_{\text{target}} = \log_2(4/3) \approx 0.415037. \quad (1.5)$$

3. **Computational Searches:** In 1990, Jeffrey M. Kubina and Marvin C. Wunderlich [5] completed an extensive Cray Y-MP search confirming zero exceptions for all $k \leq 471,600,000$.

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2 The Hermite–Padé Multi-Dimensional Framework

To investigate whether the single-variable bound $c \approx 1$ can be theoretically reduced, we consider simultaneous rational approximations around smooth rational points.

2.1 The 4th-Power Smooth Relation

Consider the identity:

$$1 - \frac{1}{81} = \frac{80}{81} = \frac{2^4 \cdot 5}{3^4} = 5 \cdot \left(\frac{2}{3} \right)^4. \quad (2.1)$$

Taking 4th roots yields:

$$\left(1 - \frac{1}{81} \right)^{1/4} = \frac{2}{3} \cdot 5^{1/4} \iff 5^{1/4} = \frac{3}{2} \left(1 - \frac{1}{81} \right)^{1/4}. \quad (2.2)$$

2.2 Asymptotic Heights and Saddle-Point Analysis

Let $f_j(z) = (1 - z)^{j/4}$ for $j \in \{1, 2, 3\}$. The Type II Hermite–Padé polynomials $Q_n(z), P_{j,n}(z)$ satisfy:

$$Q_n(z)f_j(z) - P_{j,n}(z) = z^{3n+1}R_{j,n}(z). \quad (2.3)$$

On the Riemann sphere, the saddle-point rates for $z = 1/81$ give error decay rate:

$$\mu_2 = \left(\frac{1 - \sqrt{80/81}}{1 + \sqrt{80/81}} \right)^2 = \left(\frac{9 - 4\sqrt{5}}{9 + 4\sqrt{5}} \right)^2 \approx 9.644876 \times 10^{-6}, \quad (2.4)$$

and coefficient growth rate $\mu_1 = 1/\mu_2 \approx 103,682$.

Under standard Chudnovsky–Hata transference [7, 8], simultaneous rational approximation across $d - 1 = 3$ independent linear forms divides the logarithmic height by 3:

$$c_{\text{eff}} = \frac{\ln \mu_1 + \ln D}{3(\ln(1/\mu_2) - \ln D)} \approx 0.399922 < \log_2(4/3) \approx 0.415037. \quad (2.5)$$

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3 The Critical Mathematical Obstructions

While the asymptotic exponent $c_{\text{eff}} \approx 0.3999$ appears to cross the Dickson–Pillai barrier, rigorous scrutiny reveals three profound obstacles that prevent this construction from yielding an immediate unconditional proof.

3.1 Obstruction 1: Prime-Factor Leakage Outside $S = \{2, 3, 5\}$

In attempting to decouple the algebraic power 5^m from the linear form via the Artin product formula on $K = \mathbb{Q}(5^{1/4})$, one encounters the quantity:

$$r_k = 3^k - 2^k \lfloor (3/2)^k \rfloor = 3^k \bmod 2^k. \quad (3.1)$$

For an S -unit product formula $\prod_{v \in S} |r_k|_v = 1$ to hold, r_k must have no prime divisors outside S . However, r_k is governed by 2-adic bit dynamics and exhibits arbitrary prime factors:

Proposition 3.1 (Prime-Factor Leakage). *The sequence of remainders $r_k = 3^k \bmod 2^k$ does not consist of S -units for $S = \{2, 3, 5, \infty\}$. Specifically:*

- For $k = 4$ ($m = 1$): $r_4 = 81 - 16 \cdot 5 = 1$ (trivially an S -unit).
- For $k = 8$ ($m = 2$): $r_8 = 6561 - 256 \cdot 25 = 161 = 7 \times 23$.
- For $k = 12$ ($m = 3$): $r_{12} = 531441 - 4096 \cdot 129 = 3057 = 3 \times 1019$.

Since $r_8 = 161$ is divisible by primes $7, 23 \notin S$, we have:

$$\prod_{v \in \{2, 3, 5, \infty\}} |r_8|_v = |r_8|_\infty = 161 \neq 1. \quad (3.2)$$

Accounting for arbitrary prime factors $|r_k|_p = p^{-v_p(r_k)}$ re-introduces the full difficulty of integer division into the analytic bound.

3.2 Obstruction 2: One-Dimensional Exponential Orbits vs. Subspace Transference

Simultaneous Diophantine approximation theorems (e.g., Schmidt’s Subspace Theorem) divide the height exponent by $d - 1$ when approximating a fixed vector of linearly independent numbers $(\theta_1, \dots, \theta_{d-1}) \in \mathbb{R}^{d-1}$.

However, the powers $(3/2)^k$ do not form an independent multi-dimensional target; they trace a 1-dimensional exponential trajectory. When projecting the simultaneous forms back onto the single variable $(3/2)^k$, the degree-of-freedom reduction is canceled by the logarithmic height of $5^{k/4}$, restoring the single-variable exponent barrier $c \geq 1 - \varepsilon$.

3.3 Obstruction 3: Non-Asymptotic Constant Growth

In Diophantine linear forms, the prefactor C_{final} is governed by the global zero-free radius of the Padé table and the ramification index of K . Rigorous non-asymptotic constants are typically extremely small (10^{-30} or smaller), forcing the true effective threshold K_0 into the range 10^{15} to 10^{50} . Claims of $K_0 \leq 5$ arise only when neglecting the zero-distribution prefactors.

4 Formalization and Computational Verification

4.1 Lean 4 Formal Verification (Axiom-Free)

To ensure mathematical integrity, the formalization in [‘DicksonPillai.lean’](./formalization/DicksonPillai.lean) was refactored to eliminate all unproven axioms:

- **Exact Algebraic Identities:** Proved $3^4 = 2^4 \cdot 5 + 1$ and $(3^4)^m = (2^4 \cdot 5 + 1)^m$ by definition (‘rfl’).
- **Verified Prime Leakage:** Formally proved the prime leakage counterexample $r_8 = 161 = 7 \times 23$ in Lean 4 kernel arithmetic.
- **Axiom-Free Base Verification:** Exhaustively proved that condition (1.2) holds for all $k \leq 1000$ using Lean’s native computation (‘decide’).

4.2 Multi-Core Parallel Verification

Using a 20-thread C++ bitwise streaming engine with rolling 128-bit state hashing, we have verified that zero exceptions exist up to $k = 25,000,000$, with all global minimum safety ratios occurring at $k \leq 16$.

5 Conclusion

The simultaneous Hermite–Padé framework provides valuable structural insights into the algebraic relations connecting rational powers $(3/2)^k$ to smooth number bases. However, the prime-factor leakage in $r_k = 3^k \bmod 2^k$ and the 1-dimensional nature of rational exponential orbits represent formidable mathematical barriers that prevent an elementary resolution of the Dickson–Pillai condition. Bridging the gap between $c \approx 0.999999$ and $c \leq 0.415037$ remains an open challenge at the frontier of analytic number theory.

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